Lecture 5; rates and enzymes

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Abstract. Notes for teaching Biology By The Numbers (BTN) in Spring 2026 at Rockefeller University.

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1 Significant figures

You all did great on the homework. We will return them with comments via Google Drive. One common issue to raise.

Beach season is coming. I like to ride my bike to Jacob Riis Beach on Far Rockaway. I have a fancy bike computer that measures my speed.

My avg pace is 15.221 mph and I have another ≈ 2 miles to the beach. How long will it take me? a. $15.221 \times 2 = 30.442$ min b. $15 \times 2 \approx 30$ min c. $15 \times 2 \approx f \times 10$ min

Option **a** is definitely wrong – I don't know my distance accurately enough. Options **b** or **c** are better. The choice depends on whether you have exactly 2 miles left, or you are uncertain about that number as well (e.g., you might have a few miles left).

2 Diffusion confusion – axons and timescales

Question: How long would it take a protein to diffuse the length of a 1 cm axon?[1]

a. 1 min b. 1 hour c. 1 day d. 1 week e. 1 month

Estimate

$$\tau = \frac{l^2}{6D} \approx \frac{(10^4\,\mu\mathrm{m})^2}{6 \times f \times 10\,\mu\mathrm{m}^2/\mathrm{s}} = \frac{10^8}{200}\,\mathrm{s} \approx f \times 10^5\,\mathrm{s} \times \frac{1\text{ day}}{10^5\text{ s}} \approx \mathbf{f}\text{ days}$$

Addendum: we should really treat the axon as roughly 1D since it is much longer than wide. In 1D $r^2 = 2D\tau$, giving $\tau \approx 10^6\,\mathrm{s} \approx \mathbf{10}$ days.

Julie C asks: **how do we know that 10 days is too long?** Intuitively, we compare to the timescale of neural phenomenon that depend on protein synthesis and motion. So we have to compare the timescale of the

2.1 Motor proteins to the rescue

Molecular motors operate at $\sim 1\,\mu\mathrm{m/s}$:

$$\frac{10^4\,\mu\mathrm{m}}{1\,\mu\mathrm{m/s}} = 10^4\,\mathrm{s} \approx \mathbf{3}\text{ hours}$$

Motor transport is:

- $\sim 100\times$ *slower* than diffusion across a bacterial cell ($\sim 10\,\mathrm{ms}$)
- $\sim 1000\times$ *faster* than diffusion along a 1 cm axon ($\sim 10\,\mathrm{days}$)

3 Reminder on final project

Previous meeting was BioNumbers/project discussion. We'll be posting a 1 pager about the project on the course website shortly.

[1] See the following reference for quantification of typical mouse axon lengths of ≈ 2 cm. Petkantchin R, Berchet A, Peng H, Markram H, Kanari L. Generating brain-wide connectome using synthetic axonal morphologies. Nat Commun. 2025;16(1):6611. doi:10.1038/s41467-025-62030-3

Outline for whiteboard

- sig figs (HW)
- diffusion confusion – julie c
- motor proteins
- project reminder - 1pg mention
- project example (Julie M)
- body of theseus
- grad program
- derive little's
- pools & rates (little's thm)
- RBCs
- body
- babies/yr - DIY
- steady-state
- speed clim. change
- transcription v translation
- response time
-

Some project details:

1. A biological question that begs a quantitative answer.
2. Give a 5 minute presentation explaining question+calculation. Videos on website.
3. Write a vignette addressing the question quantitatively. Website/handout/book give many examples.

You might think it's best to prepare the presentation first, but **it's actually much better to do the calculation, write the vignette, and then figure out how to present** it quickly. The writing makes it clearer in your mind. The reason we ask for the vignette after the presentation is so that you will be able to incorporate feedback.

4 Example presentation from Julie M

On self-sustaining space-ships and/or human colonies and/or planets.

5 Body of Theseus

> The ship wherein Theseus and the youth of Athens returned from Crete had thirty oars, and was preserved by the Athenians down even to the time of Demetrius Phalereus, for they took away the old planks as they decayed, putting in new and strong timber in their places, insomuch that this ship became a standing example among the philosophers, for the logical question of things that grow; one side holding that the ship remained the same, and the other contending that it was not the same.
> — Plutarch, Life of Theseus 23.1

It's said that your body replaces itself every 7 years. You can find statements like this online.

Question: How long does it take for the body to replace a cellular mass equal to our total body weight?
A. $<$ 7 yr B. $\approx$ 7 years C. $>$ 7 yr

If you start thinking about this deeply, there are cells that don't replace themselves at all, e.g., in the brain, some cardiomuscular and skeletal muscle cells, lens fiber cells. There's a nice video at NPR about these ideas.

5.1 Stocks and fluxes (Little's theorem)

The reason we are talking about this is not just because we care about the human body, but also because there is a general lesson here.

How should we **approach questions about turnover**? Here an example: I went to Berkeley for my PhD. Similar to RU, the program takes 6 yrs (avg.) and admits ≈ 35 students per year. How many students are in the program?

If you guessed $6 * 35 \approx 200$ then you applied this logic.

$$P = R \cdot T$$

$P =$ pool size (stock/inventory), $R =$ rate of arrival (flux), $T =$ turnover time or lifetime. In our example, $R = 35$ students/yr and $T = 6$ yrs so that $R \cdot T$ has units of [students]. This is called Little's Theorem.

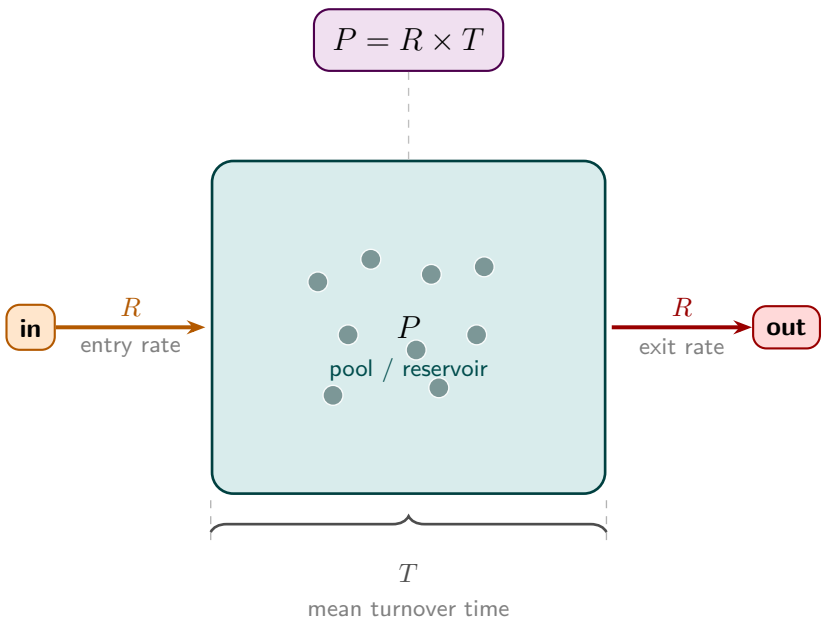


Figure 1: Illustration of Little's theorem: the mean number of items in the pool P equals the entry rate R times the mean turnover time T. Made with the help of Claude.

6 Deriving Little's theorem

Over an interval dt, $R \cdot dt$ students arrive at the PhD program. Over that same time, $T^{-1}P \cdot dt$ students leave. If P is not changing (much) then these rates are about equal.

$$T^{-1}P \cdot dt \approx R \cdot dt \implies P \approx R \cdot T$$

7 Red blood cells & body turnover

Returning to the body of theseus. Red blood cells are the most abundant cells in our body, so they are a good starting point.

Does anyone know the rate of RBC production in our bodies? i.e. the mass or RBCs produced per day? Probably you don't know this number, but we can use Little's theorem to estimate it.

- Blood volume: $\sim 5\,\mathrm{L}$; RBC fraction $\approx 50\% \Rightarrow P \approx 2\text{–}3\,\mathrm{kg}$
- Half-life: $T \approx 120\,\mathrm{days}$
- Rate: $R = P/T \approx 2500\,\mathrm{g}/120\,\mathrm{days} \approx 20\,\mathrm{g/day}$

This is a coarse estimate. Summing all tissues: $R_{\text{total}} \approx 100\,\mathrm{g/day}$, so

$$T_{\text{total}} \approx \frac{70\,\mathrm{kg}}{0.1\,\mathrm{kg/day}} \approx 700\,\mathrm{days} \approx 2\,\mathrm{years}.$$

Note: lots of variation around this average. Much detailed work to do here that is **not** an OOM estimate. See Sender et al. 2021 for detail.

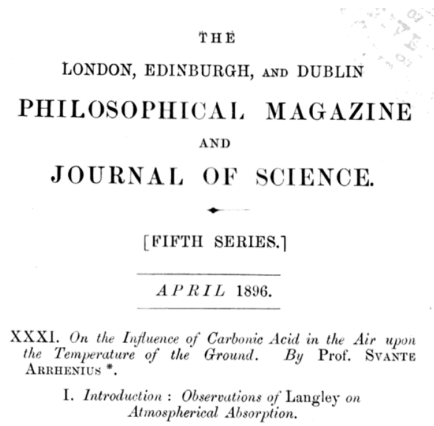
8 Global birth rate

How many babies are born each year in the world? **Try to calculate this**.

Answer: need to think in the spirit of Little's theorem.

$$P = 8 \times 10^9 \text{people}; \qquad T = 70 \text{ people} \qquad R \approx \frac{P}{T} \approx 10^8 \text{ born/year}$$

Is there anything wrong with this calculation? Little's thm assumes when the population is not changing, but of course Earth's popn is still growing. Still, if you google "births per year" the number is about 130 million, quite similar to our estimate (30% more).[2]



THE

LONDON, EDINBURGH, AND DUBLIN

PHILOSOPHICAL MAGAZINE

AND

JOURNAL OF SCIENCE.

[FIFTH SERIES.]

APRIL 1896.

XXXI. *On the Influence of Carbonic Acid in the Air upon the Temperature of the Ground.* *By* Prof. SVANTE ARRHENIUS *.

I. *Introduction: Observations of* Langley *on Atmospherical Absorption.*

Figure 2: Svante Arrhenius' 1897 paper on the effect of CO2 on climate predicted faster warming at the poles, which we have now observed.

9 Visualizing the Speed of Climate Change

We always hear about $2\,^\circ\mathrm{C}$ over some interval, like 50 or 100 years. But this is an average temperature change over the whole Earth. Things change faster over land than over the oceans and we live on the land. So we'll take a larger number, something like $3\,^\circ\mathrm{C}$ over 100 yrs. [3]

$$\approx \frac{3\,^\circ\mathrm{C}}{100 \text{ yr}}$$

Rule of Thumb: Distance from equator to poles

$d = 10^4$ km.

Interesting note, the meter was originally defined as one ten-millionth (10^{-7}) of the equator-pole distance, so 1 km is 10^{-4} times this distance (see Wiki on km).

Converting temperature change to distance (isotherms on a flat Earth):

$$\frac{10{,}000\,\mathrm{km}}{30\,^\circ\mathrm{C}} \approx 300\,\mathrm{km}/^\circ\mathrm{C}$$

$$\frac{3\,^\circ\mathrm{C}}{100\,\mathrm{yr}} \times 300\,\frac{\mathrm{km}}{^\circ\mathrm{C}} \approx \frac{1000\,\mathrm{km}}{100\,\mathrm{yr}} \approx 10\,\mathrm{km/yr} \approx 1\,\mathrm{km/month} \approx 30\,\mathrm{m/day} \approx \mathbf{1\,m/hour}$$

One way of thinking about this: you need to move 10 km/yr towards the poles. **Could a tree pull this off?** Maybe the tree can *move* by seed dispersal, but 10 km/yr is pretty far.

See: Loarie et al. (2009), "The velocity of climate change," *Nature*, doi:10.1038/nature08649. Empirical average is only $\approx 0.4\,\mathrm{km/yr}$ due to local temperature gradients and elevation effects; flat areas can reach $\approx 10\,\mathrm{km/yr}$.

10 How quickly do cells make proteins?

Can you reason about this with $P = R \cdot T$?

If the cells are growing and the daughters have the same composition as the mothers (steady-state), then we have two hand-holds.

[2] Also curious to read about changing fertility rates.

[3] Various caveats should be mentions: warming is faster at the poles and warming is not necessarily linear in time or space. The prediction of faster warming at the poles was initially due to Svante Arrhenius in 1897. Yes, the same Arrhenius responsible for the kinetic theory of reactions. His reasoning was due to the temperature dependence of water vapor in air (cold air holds less water, see the Clausius-Clayperon equation), but we now also consider the ice-albedo feedback effect.

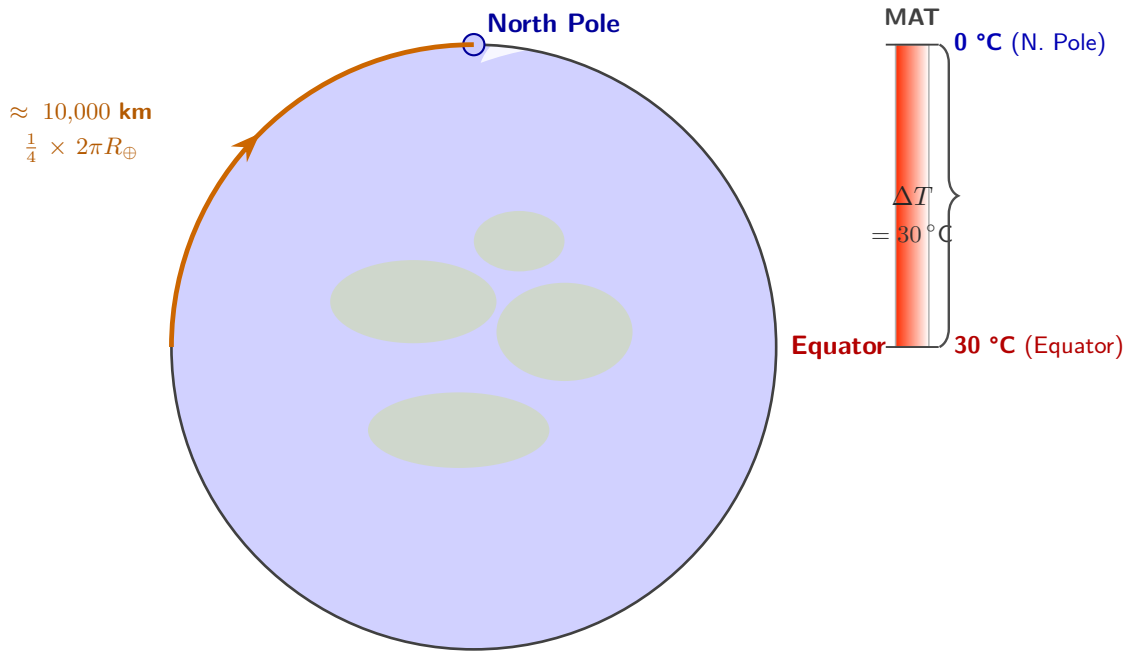


Figure 3: Pole-equator distance and mean annual temp (MAT) gradient. Made with the help of Claude.

1. rule of thumb: $f \times 10^6$ proteins/$\mu\mathrm{m}^3$.
2. turnover time T is less or equal to the division time.

since the cell must replicate all its proteins in a single division cycle.

For a bacterium like E. coli, 1 hr generations are typical. So

$$P = R \cdot T \implies T = f \times 10^6 \text{ proteins/hr/cell}$$

10.1 Larger and slower growing cells

What about mammalian cells, which are a few thousand times larger and grow more slowly? A HeLa cell has a volume of $\approx f \times 10^3$ fL and a division time around 24 hrs [BNID 103725, 109938].

The above calculation gives

$$f \times 10^6 \text{ proteins}/\mu\mathrm{m}^3 \times f \times 10^3\ \mu\mathrm{m}^3/\text{cell} \times \frac{1}{f \times 10\text{hrs}} \approx f \times 10^8 \text{ prot/hr/cell}$$

Any thoughts about this? Potential **caveats**?

The long generation times might make you wonder: **what about active degradation**?

We assumed that protein turnover was controlled by cell growth, but now there's even more time between generations and we are dealing with animal cells, which are famous for targeted protein degradation through ubiquitin and the proteasome.

If you are interested in digging into protein degradation, there is a relevant vignette from the book: How fast do RNAs and proteins degrade?

10.2 Sanity check for HeLa cells

How does the value we estimated compare to measurements? A HeLa cell contains $f \times 10^6$ ribosomes with a typical translation rate of ≈ 10 AA/s [BNID 107552, 109527].

$$\frac{f \times 10^6 \text{ ribosomes} \times 10 \text{ AA/s/ribosome}}{f \times 10^2 \text{ AA/protein}} \times f \times 10^3 \text{s/hr} \approx f \times 10^8 \text{ proteins/hr}$$

Though the two estimates agree reasonably, we made some very simple assumptions in both calculations. Here,for example, we assumed that all the ribosomes were active. Given these caveats, **the issue bears more thought**.

11 Transcription v. translation rates

Question: Which is faster **in bacteria** — transcription or translation?
a. Translation b. Transcription c. Roughly the same

Key numbers(worth memorizing, detail in Table 1):

- Translation: ~ 10–$20\,\mathrm{aa/s}$
- Transcription: ~ 40–$100\,\mathrm{nt/s}$ (≈ 10–$30\,\mathrm{codons/s}$)

Since there are 3 NT / AA in the genetic code, we see that these rates are about equal, i.e. that the rate the polymerase makes the transcript (NT/s) is similar to the rate at which the ribosome moves along it (NT/s).

$$v_{transcription} \approx f \times 10 - 10^2 \text{ NT/s}$$
$$v_{translation} \approx 10 - 20 \text{ AA/s} \times 3 \text{ NT/AA} \approx f \times 10 - 10^2 \text{ NT equiv./s}$$

Process	Bacteria	Eukaryotes	BioNumbers
Translation	~ 10–$20\,\mathrm{aa/s}$	~ 3–$10\,\mathrm{aa/s}$	[BNID 107871]
Transcription (elongation)	~ 40–$100\,\mathrm{nt/s}$	~ 30–$100\,\mathrm{nt/s}$	[BNID 111027], [BNID 105566]
Transcription (w/ pausing)		$\sim 6\,\mathrm{nt/s}$	[BNID 100661]

Table 1: Rates of transcription and translation in bacterial and eukaryotic cells.

Does this make biological sense? In bacteria, protein translation occurs simultaneous with transcription – ribosomes sit on the nascent transcripts. So translation cannot be faster than transcription (the ribosome reads the mRNA as it is being made).

In eukaryotes the per-nucleotide rates are a few-fold slower, but more importantly **introns can increase the effective transcription length (time) by a factor of** ≈ 20 [BNID 101970, 106732]. Makes you wonder about cases where Eukaryotic transcription needs to be fast, e.g., during development.

11.1 Rate of DNA replication

Does anyone **know how fast DNA polymerase is**? *E. coli* DNA polymerase II has a measured rate of ≈ 750 nt/s [BNID 104120], so it's almost 10x faster than RNA polymerase.

Interestingly, the ones we use for PCR are slower than this. E.g., my group uses q5 polymerase, which has a quoted rate of 10s / kilobase, or 100 bases/s. Any ideas why we use these slower ones (thermostability, proofreading).

12 Time to make a functional protein

Starting from an available mRNA:

$$\frac{300\,\mathrm{aa}}{10\,\mathrm{aa/s}} \approx 30\,\mathrm{s} \quad \Rightarrow \quad \text{about 1 minute}$$

Above we saw that transcribing the mRNA 'fresh' would take the same amount of time. Regardless, in prokaryotes transcription and translation are concurrent. Initiation and other steps add to the total response time.

In eukaryotes, coding-sequence translation time is similar, but:

- Absolute rates are usually a few-fold slower.
- Introns make the total transcription time tenfold longer (or more).
- Translation is cytoplasmically localized, so transcription and translation happen in serial.
- Genes involved in fast responses may be under selection for shorter introns.[4]

(The analysis above gives the time to the *first* protein; producing many copies requires additional time.)

13 Response time and turnover

13.1 mRNA Degradation and Response Times

Degradation rates set the timescale on which mRNA levels respond to induction (governed by the faster of degradation and cell-cycle dilution):

	Degradation time of mRNA	Cell-cycle time
Bacteria	$\sim 3\,\mathrm{min}$	$\sim 1\,\mathrm{h}$
Yeast	$\sim 30\,\mathrm{min}$	$\sim 3\,\mathrm{h}$
Mammalian cell line	$\sim 300\,\mathrm{min}$	$\sim 1\,\mathrm{day}$

Table 2: mRNA response time $\approx \min(\tau_{\text{deg}}, \tau_{\text{cell cycle}})$.

13.2 Protein Induction Dynamics

Following induction (e.g., IPTG addition), the protein concentration obeys:

$$\frac{dc}{dt} = \beta - \alpha\, c, \qquad \alpha = \alpha_{\text{deg}} + \alpha_{\text{dil}}$$

The steady-state level is $c_{\text{st}} = \beta/\alpha$. Starting from $c(0) = 0$:

$$c(t) = c_{\text{st}}\big(1 - e^{-\alpha t}\big) \quad \Rightarrow \quad \tau_{1/2} = \frac{\ln 2}{\alpha}$$

Three characteristic values:

- Steady-state level: $c_{\text{st}} = \beta/\alpha$
- Delay time: τ_{delay}
- Response time: $\tau_{\text{response}} \sim 1/\alpha \approx \min(\tau_{\text{deg}}, \tau_{\text{cell cycle}})$

[4] See Castillo-Davis CI, Mekhedov SL, Hartl DL, Koonin EV, Kondrashov FA. Selection for short introns in highly expressed genes. Nat Genet. 2002;31(4):415-418. doi:10.1038/ng940

Key insight: The synthesis rate β does *not* affect the response time — it only sets the final steady-state level. A higher β scales the curve up, but the time constant $1/\alpha$ is unchanged.

	Degradation time of protein	Cell-cycle time
Bacteria	hours	$\sim \mathbf{1}\,\mathrm{h}$
Yeast	$\sim 1\,\mathrm{day}$	$\sim \mathbf{3}\,\mathrm{h}$
Mammalian cell line	days	$\sim \mathbf{1}\,\mathrm{day}$

Table 3: Protein response time (bold entry dominates).

14 The rate of protein degradation by the proteasome

The above calculations should make you wonder: **how much active degradation is there in cells?**

A calculation in the book (Fig. 5) proceeds by estimating the number of proteasomes in a HeLa cell and assuming a typical rate of ≈ 5 proteins degraded / minute / proteasome complex. Notice that the proteasome is a large protein complex (long dimension $\approx f \times 10$ nm, Fig. 4). The proteasome has to be large because it degrades proteins by feeding them through a central pore on the caps (yellow). The proteins are unfolded as they are fed into a large central cavity where the peptide hydrolysis (degradation) proceeds.

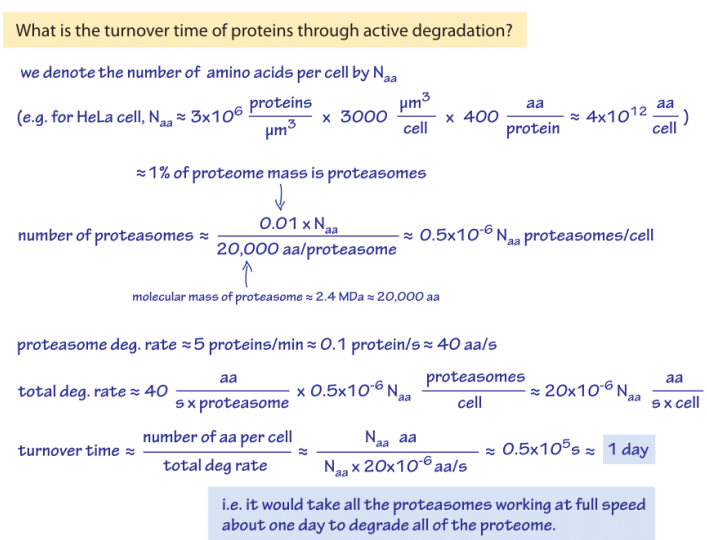


Figure 5: Estimating the protein turnover in HeLa cells that is due to proteasomes from Cell Biology by The Numbers.

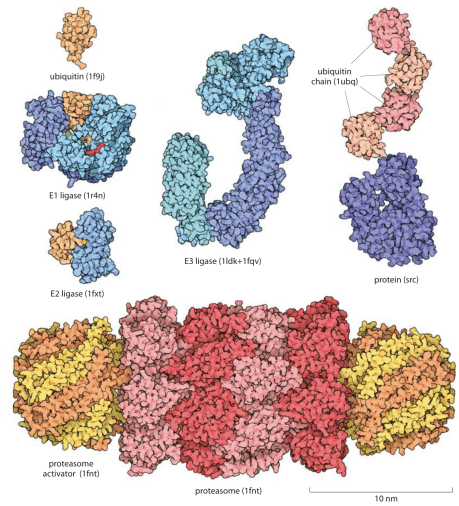


Figure 4: Illustration of the structures of the proteasome and ubiquitination / deubiquitanation machinery from Cell Biol. by the Numbers.

Any biological lessons? Based on the calculation, the turnover due to active degradation is about equal to the turnover due to dilution by cell division. That means we should double our estimate of the total translation rate. On one hand – n**ot an order of mag. change.** On the other – the **proteasome uses energy** (about 100 ATP/min/complex, [BNID 109936]) which could be a sizable increase in the energy expenditure associated with protein homeostasis (relative to a dilution only case).

Vignette: how fast does the proteasome degrade proteins?

15 Is the proteasome fast or slow?

TODO: Comparison with the diffusion limited enzyme.

16 Diffusion Across a Cell

Did this in lecture #4. Skip.

Question: How long does it take a protein to diffuse from the cell membrane to the nucleus/DNA?

Options for (bacteria, mammalian adherent cell):
$10\,\mu\mathrm{s}$ / $100\,\mu\mathrm{s}$; $10\,\mu\mathrm{s}$ / $10\,\mathrm{ms}$; $10\,\mathrm{ms}$ / $100\,\mathrm{ms}$; $10\,\mathrm{ms}$ / $10\,\mathrm{s}$; $10\,\mathrm{s}$ / $1\,\mathrm{min}$; $10\,\mathrm{s}$ / $10\,\mathrm{min}$

Diffusion Time Estimate

Root-mean-squared displacement in three dimensions:

$$r \sim \sqrt{6D\tau} \quad \Longrightarrow \quad \tau \sim \frac{r^2}{6D}$$

Cytoplasmic diffusion coefficient: $D \approx 10\,\mu\mathrm{m}^2/\mathrm{s}$.

Bacteria ($r \approx 1\,\mu\mathrm{m}$):

$$\tau \approx \frac{(1\,\mu\mathrm{m})^2}{6 \times 10\,\mu\mathrm{m}^2/\mathrm{s}} = \frac{1}{60}\,\mathrm{s} \approx 10\,\mathrm{ms}$$

Mammalian adherent cell ($r \approx 20\,\mu\mathrm{m}$):

$$\tau \approx \frac{(20\,\mu\mathrm{m})^2}{6 \times 10\,\mu\mathrm{m}^2/\mathrm{s}} = \frac{400}{60}\,\mathrm{s} \approx 10\,\mathrm{s}$$

Dimensional note: The units $\mu\mathrm{m}^2/\mathrm{s}$ naturally give the area swept by a diffusing molecule; multiplying by time gives a length scale (works cleanly in 3D).

Non-intuitive point: From a single molecule's perspective, diffusion requires no driving force and no information about concentration gradients — it is purely random thermal motion.

Further reading:

- Parry et al. (2014), "The bacterial cytoplasm has glass-like properties and is fluidized by metabolic activity," PMC3956598.
- Woodhouse & Goldstein (2015), "Hydrodynamic collective effects of active protein machines in solution and lipid bilayers," *PNAS* 112(28).

17 Can adjacent pairs of the same codon translate faster?

Skip – off topic for us.

Hypothesis: After a tRNA decodes codon i, it diffuses away and could rebind at codon $i{+}2$ (e.g., two adjacent Ser/AGU codons). Does local concentration enhancement speed up translation?

17.1 Diffusion-based estimate for translation rate

Time between decoding codon i and codon $i{+}2$:

$$\Delta\tau \approx \frac{1}{10\text{–}20\,\mathrm{aa/s}} \approx 0.1\,\mathrm{s}$$

Distance the tRNA diffuses [BNID 107935] in that time:

$$r = \sqrt{6D\,\Delta\tau} = \sqrt{6 \times 10\,\mu\mathrm{m}^2/\mathrm{s} \times 0.1\,\mathrm{s}} \approx 2\,\mu\mathrm{m}$$

This exceeds the diameter of a bacterium ($\sim 1\,\mu\mathrm{m}$) and is comparable to the radius of a yeast cell ($\sim$ 2–3 $\mu\mathrm{m}$), so the tRNA spreads over essentially the entire cellular volume.

Note: I think this suggests that the ribosome is near the diffusion limit, since time needed for tRNA turnover is similar to the diffusion time. This is worthy of further inquiry, especially since recent measurements give lower D values for tRNA in cytoplasm than those quoted above in [BNID 107935]. See Plochowietz et al. NAR 2017.

17.1.1 Copy-number argument

With [tRNA] $\approx 10\,\mu\mathrm{M}$ in a $10\,\mu\mathrm{m}^3$ yeast cell, there are $\sim 10^5$ copies of that tRNA species. One extra nearby copy is entirely negligible.

Conclusion: The local concentration enhancement is negligible; same-codon and synonymous-codon pairs translate at essentially the same rate. A 2014 follow-up study confirmed that the correlations in the original Cell paper arose from other sources.

18 Interlude: Turnover Time of Metabolites

Protein synthesis demand in yeast/*E. coli*:

$$3 \times 10^6\,\frac{\mathrm{proteins}}{\mu\mathrm{m}^3 \cdot \mathrm{h}} \times 300\,\frac{\mathrm{aa}}{\mathrm{protein}} \times \frac{4\,\mathrm{ATP}}{\mathrm{aa}} \approx 4 \times 10^9\,\frac{\mathrm{ATP}}{\mu\mathrm{m}^3 \cdot \mathrm{h}} \approx 10^6\,\frac{\mathrm{ATP}}{\mu\mathrm{m}^3 \cdot \mathrm{s}}$$

ATP concentration: $\sim 1\,\mathrm{mM} \;\rightarrow \sim 10^6\,\mathrm{ATP}/\mu\mathrm{m}^3$.

$$\tau_{\mathsf{ATP}} \sim \frac{10^6\,\mathrm{ATP}/\mu\mathrm{m}^3}{10^6\,\mathrm{ATP}/(\mu\mathrm{m}^3 \cdot \mathrm{s})} \approx \mathbf{1}\,\mathrm{s}$$

Metabolites turn over on timescales of seconds, far faster than gene-expression timescales. This motivates allosteric and post-translational regulation: metabolism cannot *wait* for gene expression.